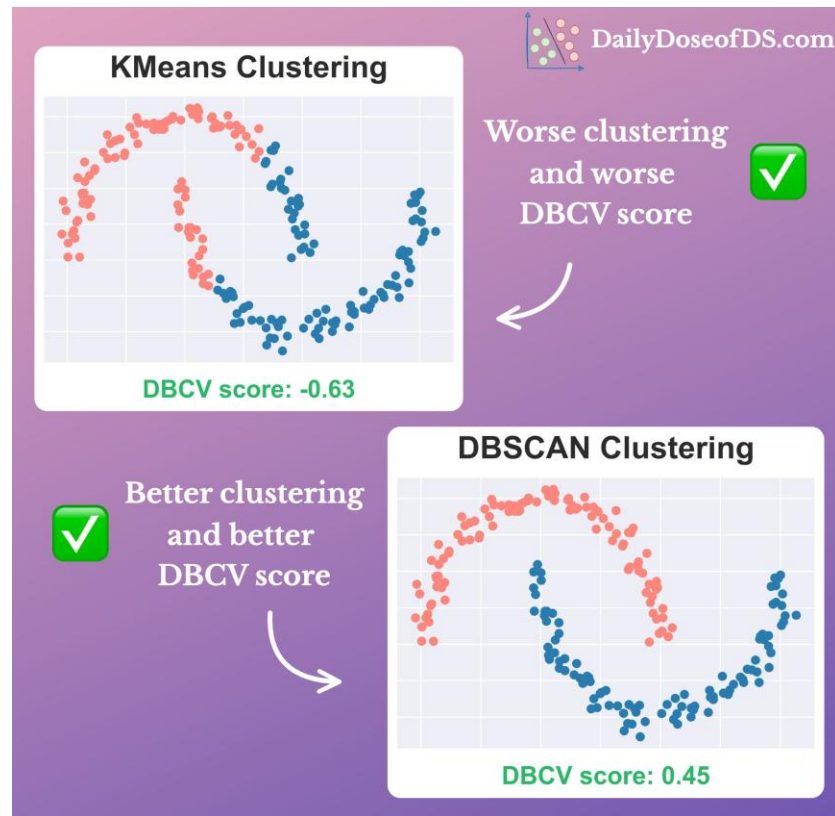
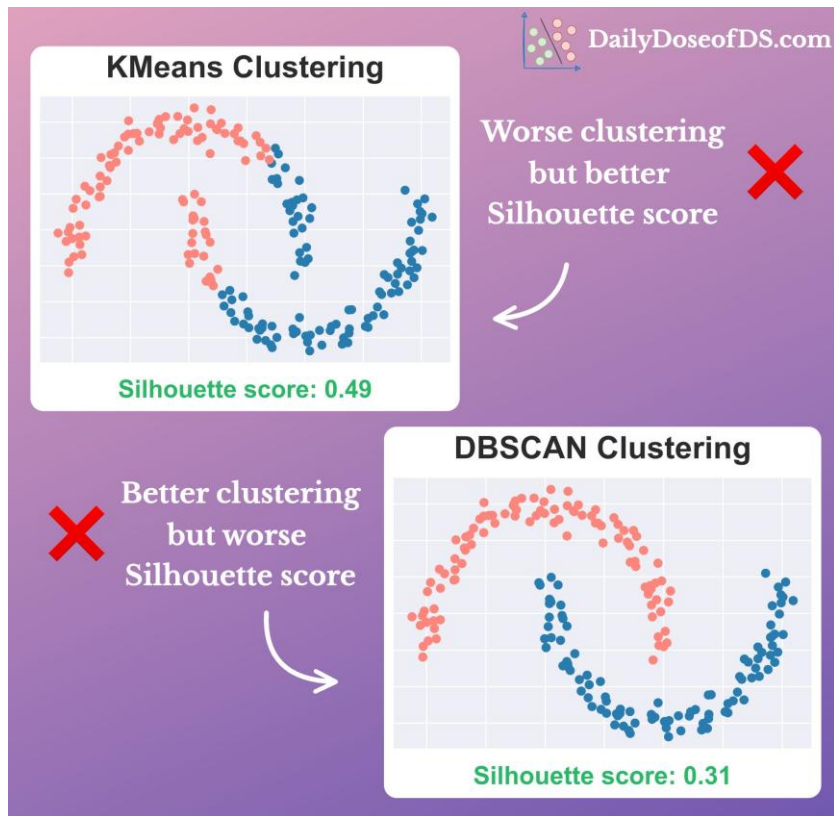


Density-Based Clustering Validation (DBCV Score)



Density-Based Clustering Validation (DBCV Score)

The Two Fundamental Values



Density Within

Measures the cohesion of points within a single cluster (DSC). We seek minimum sparseness for internal stability.



Density Between

Quantifies the separation gap between distinct clusters (DSPC). Large gaps ensure robust cluster distinctness.

$$\text{DBCV} = \sum_{i=1}^k \frac{|C_i|}{N} \cdot \text{Validity}(C_i)$$

$$\text{Validity}(C_i) = \frac{\text{DSPC}_{\min} - \text{DSC}(C_i)}{\max(\text{DSPC}_{\min}, \text{DSC}(C_i))}$$

Density-Based Clustering Validation (DBCV Score)

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Individual Cluster Quality

This formula assesses how well each cluster is formed by comparing the separation gap (DSPC) with internal sparseness (DSC).

- **DSC (Density Sparseness):** Represents the "looseness" of a cluster. Lower DSC values indicate a tightly-knit, highly dense population of points.
- **DSPC (Density Separation):** Measures the "clearance" between clusters. Higher DSPC values signify a vast, sparse region protecting clusters from overlapping
- **Numerator:** The net density separation.
- **Denominator:** Normalizes the score to a fixed range.
- **Objective:** Ensure separation outweighs internal spread.

Density-Based Clustering Validation (DBCV Score)

$$\text{DBCV} = \sum_{i=1}^k \frac{|C_i|}{N} \cdot \text{Validity}(C_i)$$

Overall System Performance

- The global score is not just an average; it is a weighted summation that favors larger, more significant clusters.
- Weighting: Factored by cluster size relative to total N.
- Robustness: Prevents small noise clusters from skewing the final result.

DBCV index values range between -1 and +1:

- +1: Strongly cohesive and well-separated clusters.
- 0: Ambiguous clustering structure.
- -1: Poorly formed clusters or incorrect assignments.

Density-Based Clustering Validation (DBCV Score)

```
!pip install hdbscan

from hdbscan.validity import validity_index

dbcv_score_dbscan = validity_index(X, dbscan_labels)
print(f"DBCV Score_dbscan: {dbcv_score_dbscan:.4f}")

dbcv_score_kmeans = validity_index(X, kmeans_labels)
print(f"DBCV Score_kmeans: {dbcv_score_kmeans:.4f}")
```